SIMULINK MODEL TECHNICAL REPORT

Hydroelectric Power Plant Digital Twin | Kavşak Bendi HEPP Reference

Project	Prepared by	Model scope	Date
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Abstract

This report documents the supplied HPP_Digital_Twin_Model.slx file and the accompanying Simulink screenshots. The current model is broader than the initial hydraulic-power-only diagram: it contains hydraulic/reservoir, turbine-generator, transformer/grid, and protection subsystems. The turbine-generator conversion chain applies turbine and generator efficiencies and limits electrical output to the reference installed capacity.

1. Model architecture

The top-level Simulink model contains the following subsystems and monitoring elements:

Subsystem / element	Purpose
Hydraulics_and_Reservoir	Represents inflow/outflow, reservoir volume/level, net head, and hydraulic power signals.
Turbine_Generator_System	Converts hydraulic power to electrical power using turbine and generator efficiency gains and a saturation limit.
Transformer_and_Grid	Represents the transformer/grid interface and calculates an illustrative grid-side current signal.
Protection_System	Compares high-voltage current against an illustrative threshold and produces a trip signal.
Scopes / Display / signal blocks	Provide visual monitoring of selected model variables and outputs.

2. Turbine-generator conversion chain

The inspected turbine-generator subsystem has one hydraulic-power input, two Gain blocks, a Saturation block, and one electrical-power output. The block parameters are:

Block	Parameter	Meaning
Gain	0.90	Assumed turbine efficiency
Gain1	0.96	Assumed generator efficiency
Saturation	Lower = 0; Upper = 191.28	Limits output to the reference installed capacity (MW)

The conversion is represented by $P_m = P_h \times \eta_t$ and $P_e = P_m \times \eta_g$, followed by $0 \le P_e \le 191.28$ MW. The efficiency values are simplified model assumptions, not verified operating curves.

3. Hydraulic power reference calculation

The hydraulic-power equation is $P_h = \rho \times g \times Q \times H_n$. For $\rho = 1000$ kg/m³, $g = 9.81$ m/s², $Q = 237.55$ m³/s, and $H_n = 95$ m:

$$P_h \approx 221.39 \text{ MW}$$

After applying $\eta_t = 0.90$ and $\eta_g = 0.96$, the estimated electrical output is approximately 191.28 MW, consistent with the 191.28 MW saturation limit. The displayed value is rounded.

4. Supplied model screenshots

Figure 1 shows the hydraulic-power calculation block chain. Figure 2 shows the conversion chain including turbine and generator efficiencies.

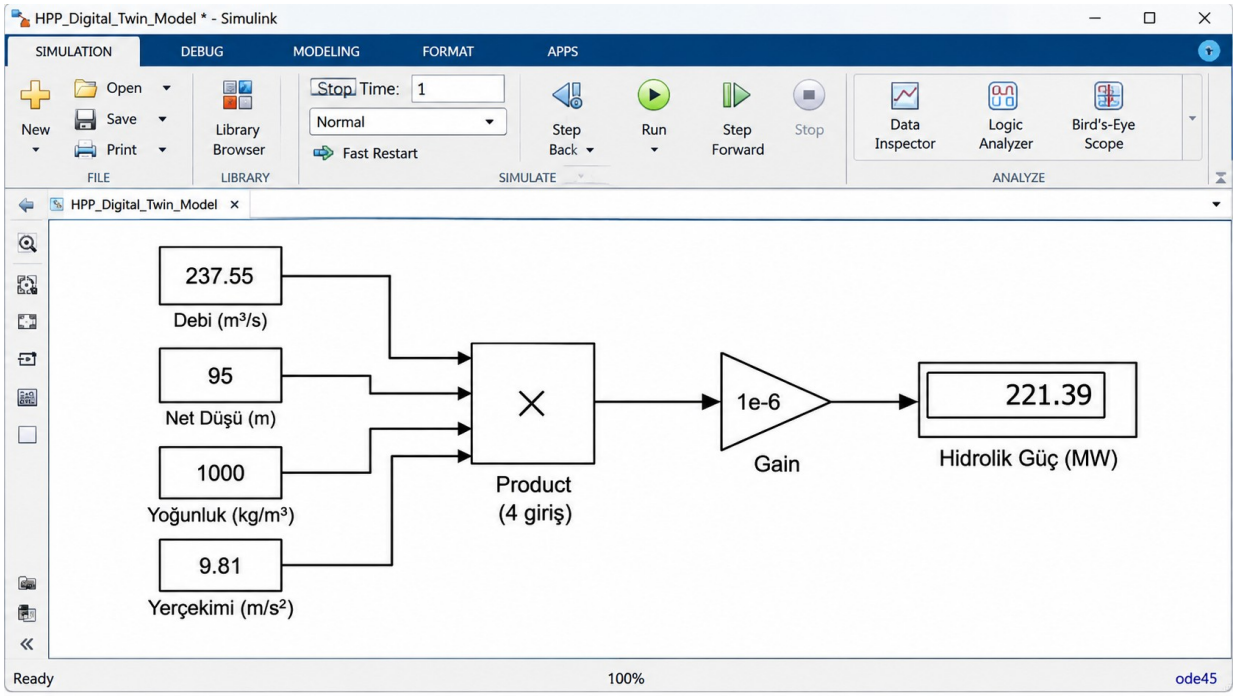


Figure 1. Hydraulic power calculation: four inputs → Product → 1e-6 Gain → Display.

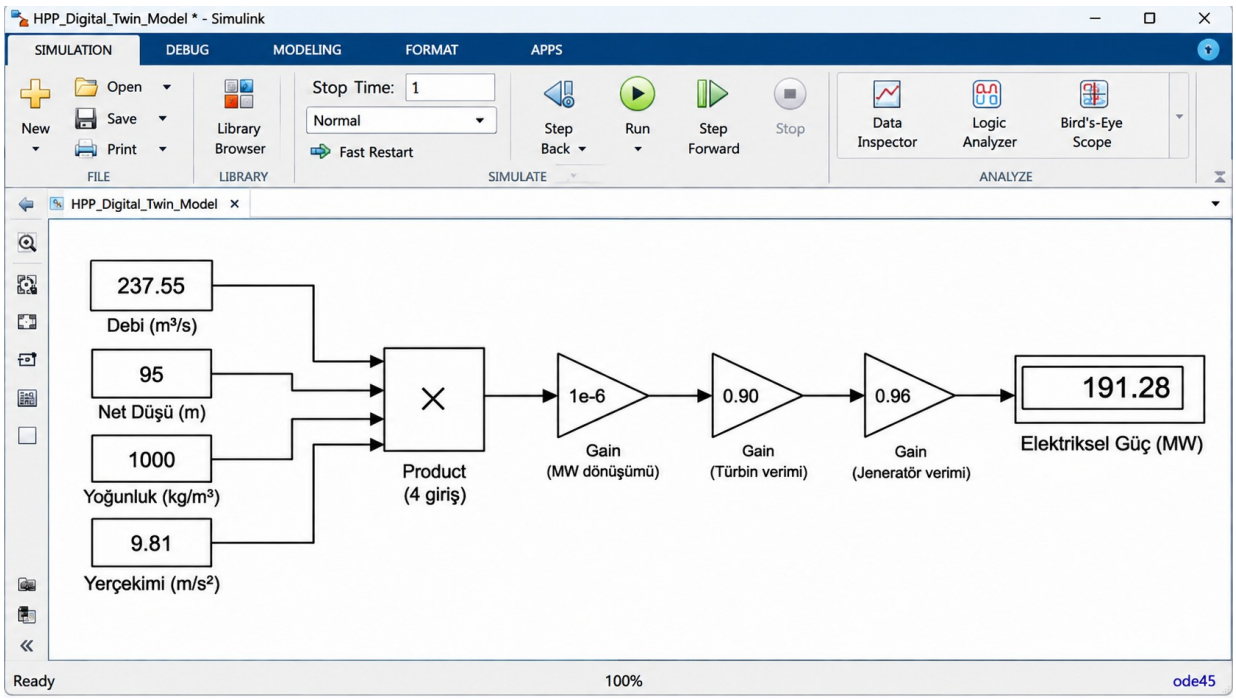


Figure 2. Electrical power conversion with efficiency gains and installed-capacity limit.

5. Transformer, grid, and protection representation

The transformer/grid subsystem includes a 0.995 gain and a high-voltage current calculation based on three-phase apparent power, 154 kV, and 0.90 power factor. The protection subsystem compares high-voltage current against an illustrative 850 A threshold. This is a simplified educational demonstration; it is not a protection coordination study and the threshold is not a real relay setting.

6. Assumptions and limitations

- The model uses reference and assumed inputs; it is not connected to live SCADA or plant sensors.
- The efficiencies 0.90 and 0.96 are fixed values and do not represent full turbine or generator performance maps.
- Reservoir dynamics are simplified and do not include detailed waterway transients or water hammer.
- The transformer/grid and protection subsystems are simplified representations, not validated plant single-line or relay models.
- The supplied screenshots and model file are retained as provided; the model has not been simulated in this packaging environment.

7. Running the model

Open Simulink/HPP_Digital_Twin_Model.slx in MATLAB/Simulink. Inspect the four main subsystems, confirm block parameters, and run the simulation. MATLAB/Simulink R2023b or a compatible later release is recommended. Compatibility adjustments may be required in other releases.

Conclusion

The supplied Simulink file represents a multi-subsystem educational model of the hydropower energy chain, extending from hydraulic/reservoir variables to turbine-generator conversion, transformer/grid interface, and illustrative protection logic. Further validation against verified plant data and detailed equipment models is required before any engineering or operational use.